

Name
Date
Course

Solving Nonlinear Inequalities (Polynomials) Homework

Solve. Write the solutions graphically and in interval notation.

① $-x^2 > 17x + 72$

② $49 - x^2 \leq 0$

③ $x^3 \geq x$

④ $3x^3 + 21x^2 + 36x < 0$

⑤ $4x^4 < 325x^2 - 81$

⑥ $x^2 < 2x - 1$

$$\textcircled{7} \quad 9x^2 + 30x + 25 > 0$$

$$\textcircled{8} \quad x^3 - 8 \leq 6x - 3x^2$$

$$\textcircled{9} \quad 3x^3 + 7x^2 \geq 192x + 448$$

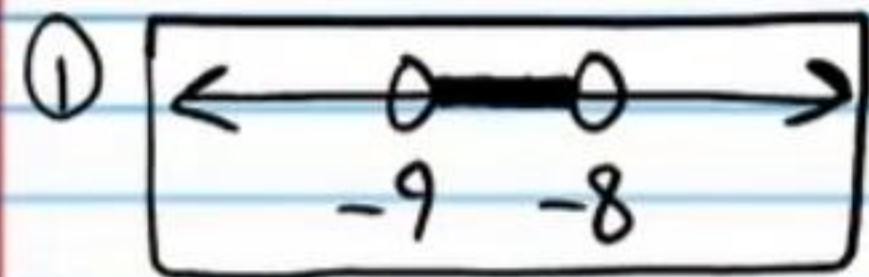
$$\textcircled{10} \quad x^2 \leq 6$$

$$\textcircled{11} \quad 4x < x^2 + 1$$

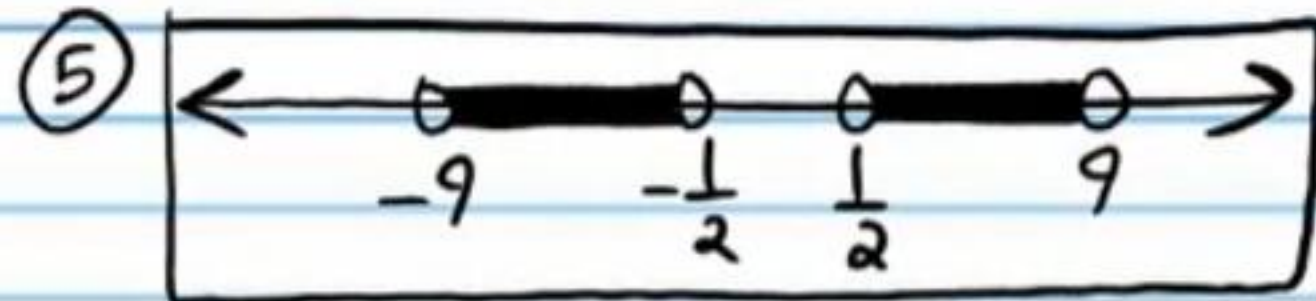
$$\textcircled{12} \quad x^3 + 1 > 0$$

$$\textcircled{13} \quad 64x^3 \leq 125$$

Answers



$$(-9, -8)$$

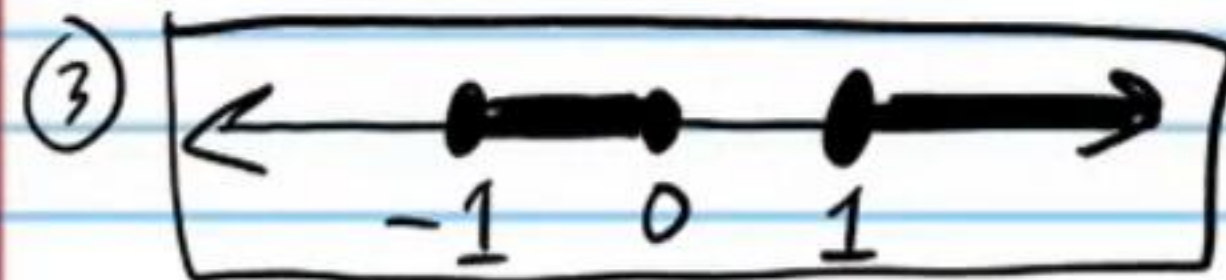


$$(-9, -\frac{1}{2}) \cup (\frac{1}{2}, 9)$$

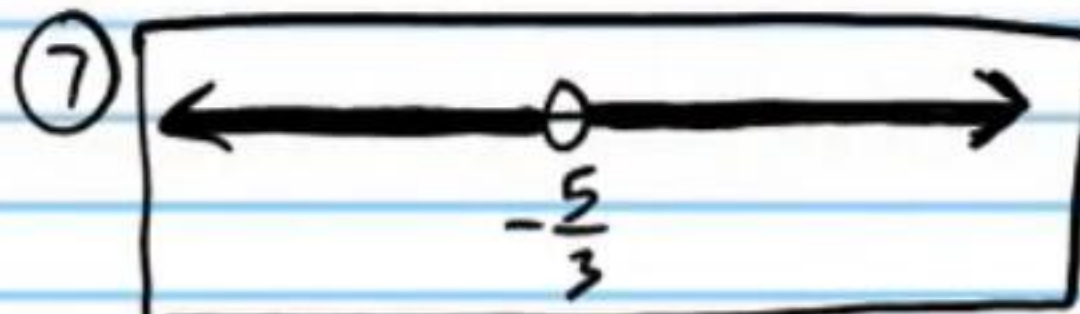


$$(-\infty, -7] \cup [7, \infty)$$

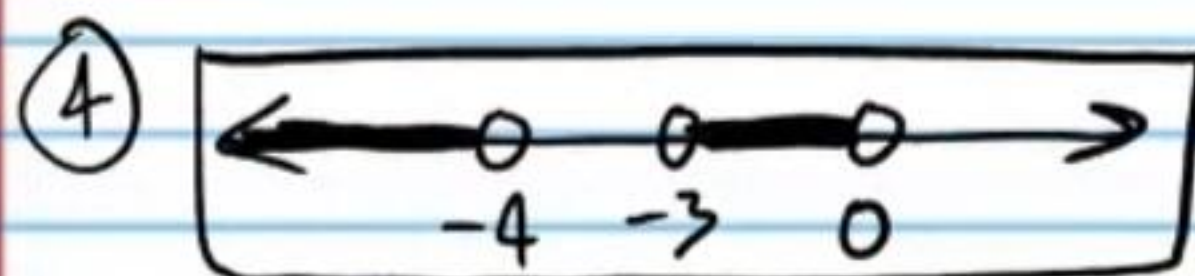
⑥ No Solution



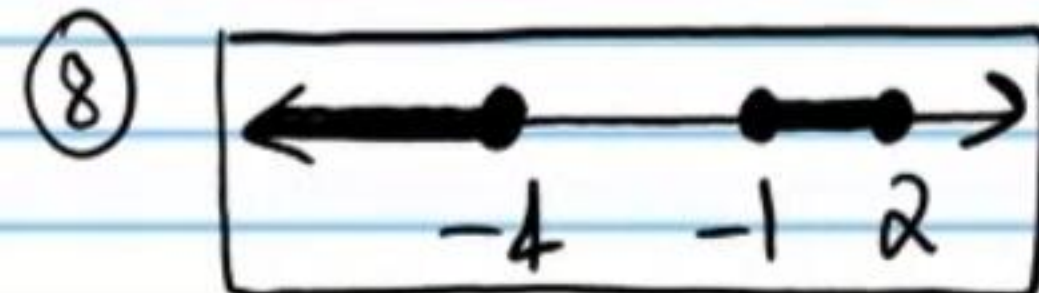
$$[-1, 0] \cup [0, 1]$$



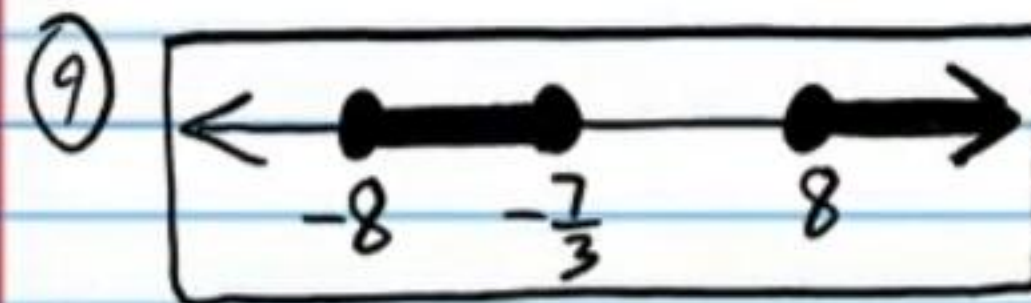
$$(-\infty, -\frac{5}{3}) \cup (-\frac{5}{3}, \infty)$$



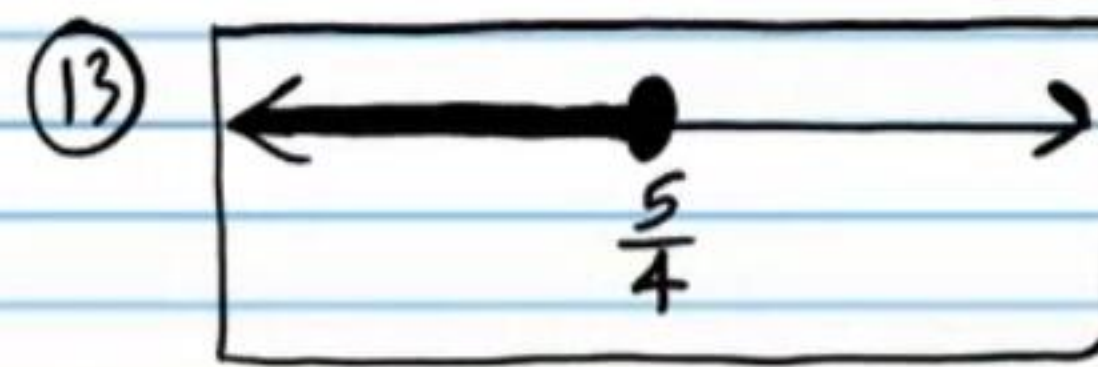
$$(-\infty, -4) \cup (-3, 0)$$



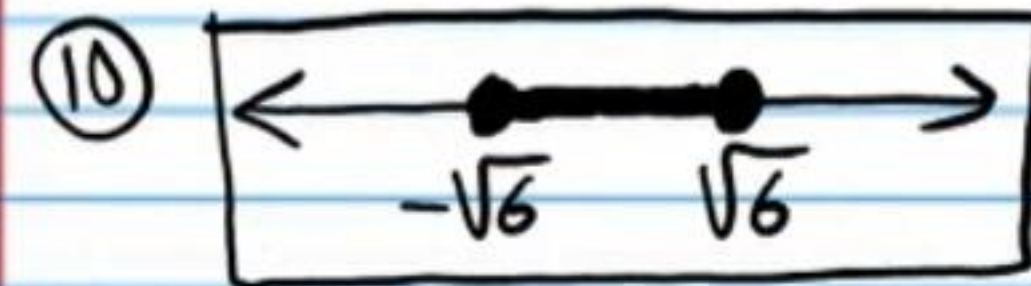
$$(-\infty, -4] \cup [-1, 2]$$



$$[-8, -\frac{7}{3}] \cup [8, \infty)$$



$$(-\infty, \frac{5}{4}]$$



$$[-\sqrt{6}, \sqrt{6}]$$



$$(-\infty, 2-\sqrt{3}) \cup (2+\sqrt{3}, \infty)$$



$$(-1, \infty)$$